

CELIA RUBIO-MADRIGAL

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 LinkedIn

 Google Scholar

I study machine learning on structured data, with a focus on how input structures shape both data representation and computation. I am IT at EGMO, a "la Caixa" Fellowship recipient, and attended the 9th and 13th HLF.

EXPERIENCE



- **Doctoral researcher** – CISPA Helmholtz Center for Information Security **2023-now**
– Graph learning, generalization, and structure-aware ML – Advisor: Dr. Rebekka Burkholz 
- **RA: Departmental collaboration** – Ministerio de Educación & Universidad Complutense de Madrid **2021-22**
– "Identification of Boolean functions that need big circuits"  – Advisor: Dr. Ismael Rodríguez Laguna 
- **REU: Introduction to research ICMAT-Severo Ochoa** – Instituto de Ciencias Matemáticas (ICMAT) **2021**
– "Differential geometry methods in ML applications"  – Advisor: Dr. David Martín de Diego 
- **Consultant Developer intern** – Guidewire Software – Design of a search engine to find raised bugs. **2021**

EDUCATION



- **PhD in Computer Science** – CISPA & Universität des Saarlandes (Saarbrücken, Germany) **2023-now**
- **MSc in Advanced CS with AI** – University of Strathclyde (Glasgow, UK) – 8.57/10 (Distinction) **2022-23**
– Recipient of the "la Caixa" Fellowship  – Department's Best Overall Performance Prize (Top 1)
- **BSc in Mathematics** – Universidad Complutense de Madrid (Spain) – 9.08/10 **2017-22**
- **BSc in Computer Science** – Universidad Complutense de Madrid (Spain) – 9.16/10 **2017-22**

PUBLICATIONS



- C.R.M., R. Burkholz. *Fixed Aggregation Features Can Rival GNNs*. ICML'26. 
- A.G. Reddy, R. Verma, C.R.M., K. Muandet, R. Burkholz. *Boosting for Predictive Sufficiency*. ICLR'26. 
- A.G. Reddy, C.R.M., R. Burkholz, K. Muandet. *When Shift Happens - Confounding Is to Blame*. ICLR'26. 
- T. Jacobs, A. Gadhikar, C.R.M., R. Burkholz. *Hyperbolic Aware Minimization: Implicit Bias for Sparsity*. ICLR'26. 
- C.R.M. *, A. Jamadandi *, R. Burkholz. *GNNs Getting ComFy: Community and Feature Similarity Guided Rewiring*. ICLR'25.  * Equal contribution.
- A. Jamadandi *, C.R.M. *, R. Burkholz. *Spectral Graph Pruning Against Over-Squashing and Over-Smoothing*. NeurIPS'24.  * Equal contribution.

AWARDS



- **Postgraduate**
- 13th Heidelberg Laureate Forum: 200 international young researchers network with math and CS laureates **2026**
- Dept. Best Overall Performance Prize: University of Strathclyde's top student in MSc Advanced CS **2023**
-  Nova III Student List: Spain's top 10 talents under 25 in Computer Science **2023**
- 9th Heidelberg Laureate Forum. Featured in:  HLFF Spotlight,  Tribuna Complutense **2022**
-  "la Caixa" Fellowship: Full funding for MSc studies. Featured in:  El País,  El Plural,  TeleMadrid **2022**

• Undergraduate

- 📄 Aula de Verano “Blas Cabrera” for Spain’s top undergrad records: *Intl. Univ. Menéndez Pelayo (UIIMP)* 2022
- 📄 SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* 2022
- Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* 2018-22

• Pre-undergraduate

- 📄 EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* 2017
- 📄 Premio Extraordinario de Bachillerato: *Madrid's top 25 in general-content exam (1000€)* 2017

TALKS



- **Invited talks** – *Seminars and reading groups* 2025-26
- 📄 GLOW reading group, 📄 TUWien ML RU seminar, 📄 Cohere For AI reading group, CISPA ML seminar.
- **Contributed talks** – *Conferences and workshops* 2024-25
- Lightning talks: GTML 📄 2025, MLG 📄 2025 (2), NetSci 📄 2025. Contributed talk: HAICON 📄 2024.
- **Non-archival posters** – *Non-archival conferences, workshops, and sessions* 2024-26
- GSP 📄 2026 (2), CPAL 📄 2026 (3), WiML 📄 2024 📄 2025, ELLIS Pre-NeurIPS 📄 2024 📄 2025.

TEACHING



- **Math Olympiad Trainer** – 📄 *International EGMO Preparatory Seminar in Bonn, Germany* 2026
- **Master's and Bachelor's thesis advising** – *CISPA & Universität des Saarlandes (3)* 2024-now
- **Teaching Assistant: Laboratory demonstrator (7h/week)** – *University of Strathclyde* 2023
- “CS310 Foundations of AI” for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel 📄

SERVICE



- **Academic service** 2024-now
- Reviewer for MLG @ ECML PKDD 2025, ICLR 2026 – Top 200 Reviewers (1%), CPAL 2026, ICML 2026.
- Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.
- **Math Olympiad service**
- 📄 IT at European Girls’ Mathematical Olympiad (EGMO) with Joseph Samuel Myers 📄 2023-now
- Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. 2021-22
- Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid. 2020-24
- Guide at the Iberoamerican Mathematical Olympiad 2018.
- **Student representation** – *Facultad de Informática (Fdi), Universidad Complutense de Madrid* 2020-22
- Secretary of the Fdi Student Council 2021-22. Member of the Fdi Faculty Board 2020-22.
- **L^AT_EX package development: ddphonism** 2019
- 📄 Twelve-tone music notation diagrams with TikZ. Included in MiKTeX and TeXLive (try it in Overleaf)
- **Mentoring and outreach**
- Mentor for new “la Caixa” Fellows 2023-now
- Outreach articles for DivulgaMAT (RSME journal): *Serialismo y matemáticas* (i-iv) 📄 📄 📄 📄 2019
- Outreach videos for the Spanish math education community: 📄 📄 📄 2018-21

LANGUAGES



Spanish – Native

English – TOEFL iBT 110/120 (2021)

German – Goethe-Zertifikat B1 (2017)